WORKED SOLUTION

STEP II 2015, Question 6

Pure • Integration by Symmetry

STEP II 2015, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q6

The key idea

Writing $\sin x$ as $\cos\left(\frac{1}{2}\pi - x\right)$ turns $1 + \sin x$ into a double-angle expression, and then $\sec^2$ integrates at once. The substitution $y = \pi - x$ reflects $[0, \pi]$ about its midpoint, and $\sin x$ doesn't notice the reflection. So any polynomial factor in front of $f(\sin x)$ can be traded for its reflected version. In part (iii) the cubic has been chosen so that this trade gets rid of every power of x at once.

Solution

Part (i)

Since $\sin x = \cos\left(\frac{1}{2}\pi - x\right)$ and $1 + \cos 2\theta = 2\cos^2\theta$, taking $\theta = \frac{1}{4}\pi - \frac{1}{2}x$ gives

$$1 + \sin x = 1 + \cos\left(\tfrac{1}{2}\pi - x\right) = 2\cos^2\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right).$$

Taking reciprocals and multiplying by 2 gives $\sec^2\left(\frac{1}{4}\pi - \frac{1}{2}x\right) = \dfrac{2}{1 + \sin x}$, as required. (Both sides are undefined at the same points, where $\sin x = -1$.)

Hence

$$\int \frac{1}{1 + \sin x}\,\mathrm{d}x = \frac{1}{2}\int \sec^2\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right)\mathrm{d}x = -\tan\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right) + C,$$

because differentiating $\tan\left(\frac{1}{4}\pi - \frac{1}{2}x\right)$ produces a factor of $-\frac{1}{2}$ by the chain rule.

Part (ii)

Let $I = \int_0^\pi x\,\mathrm{f}(\sin x)\,\mathrm{d}x$. Put $y = \pi - x$, so $\mathrm{d}x = -\mathrm{d}y$, the limits $x = 0, \pi$ become $y = \pi, 0$, and $\sin x = \sin(\pi - y) = \sin y$. Then

$$\begin{aligned} I &= \int_\pi^0 (\pi - y)\,\mathrm{f}(\sin y)\,(-\mathrm{d}y) = \int_0^\pi (\pi - y)\,\mathrm{f}(\sin y)\,\mathrm{d}y \\ &= \pi \int_0^\pi \mathrm{f}(\sin y)\,\mathrm{d}y - \int_0^\pi y\,\mathrm{f}(\sin y)\,\mathrm{d}y = \pi \int_0^\pi \mathrm{f}(\sin x)\,\mathrm{d}x - I, \end{aligned}$$

since y is only a dummy variable. So $2I = \pi \int_0^\pi \mathrm{f}(\sin x)\,\mathrm{d}x$, which is the result.

Ask yourself The unknown integral I has appeared on both sides. Is that a problem?

No. It's exactly what we want. The substitution doesn't evaluate I; it gives us a linear equation for I, and we solve that.

Now take $\mathrm{f}(t) = \dfrac{1}{1+t}$, which is continuous for $t \in [0, 1]$, the values $\sin x$ takes on $[0, \pi]$. By part (i),

$$\int_0^\pi \frac{1}{1+\sin x}\,\mathrm{d}x = \Big[-\tan\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right) \Big]_0^\pi = -\tan\left(-\tfrac{1}{4}\pi\right) + \tan\tfrac{1}{4}\pi = 1 + 1 = 2,$$

and $\tan\left(\frac{1}{4}\pi - \frac{1}{2}x\right)$ is continuous on $[0, \pi]$, because the argument stays in $[-\frac{1}{4}\pi, \frac{1}{4}\pi]$. Therefore

$$\int_0^\pi \frac{x}{1+\sin x}\,\mathrm{d}x = \frac{\pi}{2} \times 2 = \pi.$$

Part (iii)

Use the same reflection. Let $g(x) = 2x^3 - 3\pi x^2$. Expanding,

$$\begin{aligned} g(\pi - x) &= 2(\pi^3 - 3\pi^2 x + 3\pi x^2 - x^3) - 3\pi(\pi^2 - 2\pi x + x^2) \\ &= -\pi^3 + 3\pi x^2 - 2x^3 = -\pi^3 - g(x). \end{aligned}$$

Let $J = \int_0^\pi \dfrac{g(x)}{(1+\sin x)^2}\,\mathrm{d}x$. The substitution $y = \pi - x$, exactly as in part (ii), gives

$$J = \int_0^\pi \frac{g(\pi - y)}{(1+\sin y)^2}\,\mathrm{d}y = -\pi^3 K - J, \qquad \text{where } K = \int_0^\pi \frac{\mathrm{d}x}{(1+\sin x)^2}.$$

So $J = -\frac{1}{2}\pi^3 K$.

Ask yourself Why does this particular cubic work so well?

In terms of $u = x - \frac{1}{2}\pi$, the distance from the midpoint, $g(x) = 2u^3 - \frac{3}{2}\pi^2 u - \frac{1}{2}\pi^3$. Since $\sin x = \cos u$, the function $1/(1+\sin x)^2$ is even in u, so the odd terms $2u^3$ and

$-\frac{3}{2}\pi^2 u$ integrate to zero over the symmetric interval. Only the constant $-\frac{1}{2}\pi^3$ is left, which is what we found. Part (ii) is the same thing with $x = u + \frac{1}{2}\pi$.

To find K, square the identity from part (i): $\dfrac{1}{(1+\sin x)^2} = \frac{1}{4}\sec^4\left(\frac{1}{4}\pi - \frac{1}{2}x\right)$. Substitute $u = \frac{1}{4}\pi - \frac{1}{2}x$, so $\mathrm{d}x = -2\,\mathrm{d}u$, and $x = 0, \pi$ become $u = \frac{1}{4}\pi, -\frac{1}{4}\pi$. Writing $\sec^4 u = (1+\tan^2 u)\sec^2 u$, whose integral is $\tan u + \frac{1}{3}\tan^3 u$,

$$K = \int_{\pi/4}^{-\pi/4} \tfrac{1}{4}\sec^4 u\,(-2\,\mathrm{d}u) = \frac{1}{2}\int_{-\pi/4}^{\pi/4} (1+\tan^2 u)\sec^2 u\,\mathrm{d}u$$
$$= \frac{1}{2}\Big[\tan u + \tfrac{1}{3}\tan^3 u\Big]_{-\pi/4}^{\pi/4} = \frac{1}{2}\left(\tfrac{4}{3} + \tfrac{4}{3}\right) = \frac{4}{3}.$$

Therefore

$$\int_0^{\pi} \frac{2x^3 - 3\pi x^2}{(1+\sin x)^2}\,\mathrm{d}x = -\frac{\pi^3}{2}\cdot\frac{4}{3} = -\frac{2\pi^3}{3}.$$

Checks

- Differentiate $-\tan\left(\frac{1}{4}\pi - \frac{1}{2}x\right)$ and you get $\frac{1}{2}\sec^2\left(\frac{1}{4}\pi - \frac{1}{2}x\right) = 1/(1+\sin x)$.
- Signs: in part (ii) the integrand is positive, and $\pi \approx 3.14$ is believable, since $\frac{1}{2} \le 1/(1+\sin x) \le 1$ gives bounds of $\frac{1}{4}\pi^2 \approx 2.47$ and $\frac{1}{2}\pi^2 \approx 4.93$. In part (iii), $2x^3 - 3\pi x^2 = x^2(2x - 3\pi) < 0$ on $(0, \pi]$, so a negative answer is right.
- Numerical quadrature gives $\int_0^{\pi}(1+\sin x)^{-2}\,\mathrm{d}x \approx 1.33333$ and the part (iii) integral ≈ -20.6709, which agree with $\frac{4}{3}$ and $-\frac{2}{3}\pi^3$.

Common slips

- Losing the factor $-\frac{1}{2}$ from the chain rule in part (i), and getting $+2\tan(\dots)$ or $-\frac{1}{2}\tan(\dots)$.
- Forgetting that swapping the limits cancels the minus sign from $\mathrm{d}x = -\mathrm{d}y$.
- Trying integration by parts three times in part (iii). It can be done, but the reflection does it in one line.

Beyond STEP

The trick in part (ii) uses symmetry. If a weight function is unchanged by a reflection, only the part of the integrand that is even about the centre of reflection contributes. You'll meet this idea all over first-year mathematics: in Fourier series, odd functions have no cosine terms, and in probability, a distribution symmetric about μ has its mean at μ. The substitution $t = \tan\frac{1}{2}x$ is a systematic alternative for any rational function of $\sin x$ and $\cos x$. For more integrals like these, see Siklos, *Advanced Problems in Mathematics*.

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